# Practice workbook · Normal, boundary and erroneous test data

Choosing, labelling and contrasting normal, boundary and erroneous test data for a stated rule or program.

6 question parts: 1 worked-example retry and 5 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: CSC-T2-09

# 1. Question 2 · part c.i · 2 marks

2025 JUN · Paper 02

Worked-example retry · Attempt part c.i only.

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**2** Programs can process numerical data.

(a) Identify the method used to check input data matches set rules.

**(1)**

- ☒ **A** Abstraction
- ☒ **B** Encryption
- ☒ **C** Iteration
- ☒ **D** Validation

(b) **Figure 2** shows a section from an employee payslip produced by a computer program.

| Pay Rate | Hours | | Total Pay |
|---|---|---|---|
| £ 18.75 | 19 | | £ 356.25 |
| | Tax | 20% | £ 71.25 |
| | | Income | £ 285.00 |

**Figure 2**

(i) Complete the table to give **one** data type shown in **Figure 2** and give an example value.

**(2)**

| | |
|---|---|
| **Data type** | |
| **Example** | |

(ii) Complete the table to give an input, a process and an output used in the program to produce the payslip.

**(3)**

| | |
|---|---|
| **Input** | |
| **Process** | |
| **Output** | |

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# 1. Question 2 · part c.i · 2 marks

2025 JUN · Paper 02

Worked-example retry · Attempt part c.i only.

(c) Numerical values can be represented in denary, binary and hexadecimal.

A program converts denary values in the range 0 to 255 into hexadecimal.

(i) The program is tested using appropriate integer values.

Complete the table to show an example of normal, boundary and erroneous test data for this program.

(2)

| | Test data |
|---|---|
| **Normal** | |
| **Boundary** | |
| **Erroneous** | |

(ii) The program uses subprograms to do the conversion.

Open **Q02cii** in the code editor.

Amend the code to complete the program to:

- convert the user input to integer
- check the input is between 0 and 255
- call the **getHex** subprogram, with the user input as an argument.

Do not add any further functionality.

Save your code as **Q02ciiFINISHED** with the correct file extension for the programming language.

(4)

**(Total for Question 2 = 12 marks)**

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## 1. Question 2 · part c.i · 2 marks · Working and answer

Working and answer

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## 2. Question 2 · part c · 3 marks

2021 JUN · Paper 02

Further practice · Attempt part c only.

**2** A book club uses computer applications.

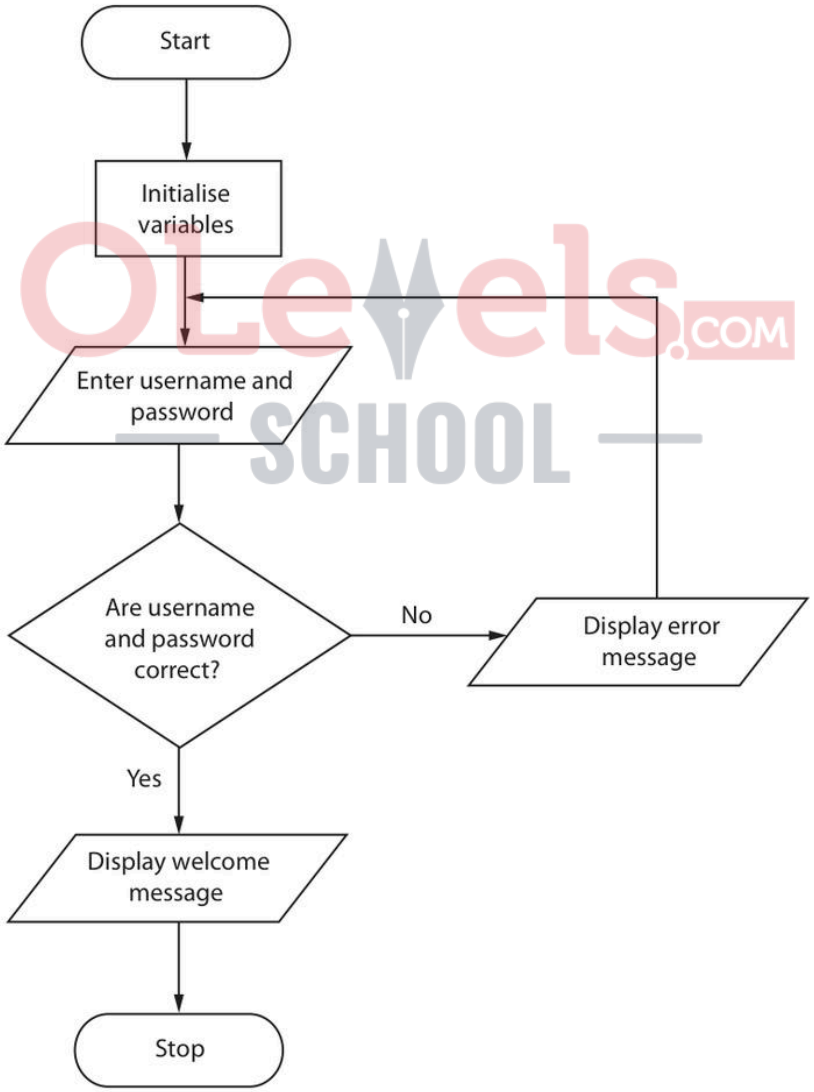
(a) The club wants a program to ensure that logins are valid.

A flowchart for a test version of the program has been designed.

The test version uses:

- a username of bard423
- a password of nX2934?

Here is the flowchart.




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## 2. Question 2 · part c · 3 marks

2021 JUN · Paper 02

Further practice · Attempt part c only.

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Write a program to implement the logic in the flowchart.

Open **Q02a** in the code editor.

You must use the structure given in **Q02a** to write the program.

Do not add any further functionality.

Save your code as **Q02aFINISHED** with the correct file extension for the programming language.

**(11)**

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# 2. Question 2 · part c · 3 marks

2021 JUN · Paper 02

Further practice · Attempt part c only.

(b) The club sells books.

It needs a computer program to monitor:

- the number of books sold
- the amount of profit made.

**Q02b** provides a structure for the program.

Open **Q02b** in the code editor.

Amend the code to complete the If statement used to produce the outputs described in the table.

| Condition | Output message |
|---|---|
| Number of books sold is at least 5 and profit made is at least 10 | Sales and profit are good this week |
| Number of books sold is over 20 and profit made is at least 20 | Sales and profit are excellent this week |
| Number of books sold is under 5 or profit made is under 5 | Poor performance this week |
| All other inputs | Alert manager |

Do not add any further functionality.

Save your code as **Q02bFINISHED** with the correct file extension for the programming language.

(4)

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## 2. Question 2 · part c · 3 marks

2021 JUN · Paper 02

Further practice · Attempt part c only.

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(c) Boundary testing is an important part of program development.

The boundary value is the most extreme value that will pass the test.

For example, if the boundary condition is no more than 30 then the boundary test data would be 30.

Complete the test table to show the boundary test data that could be used for each test.

(3)

| | Test data | Expected output |
|---|---|---|
| booksSold | | Poor performances this week |
| profit | | |
| booksSold | | Sales and profit are good this week |
| profit | | |
| booksSold | | Sales and profit are excellent this week |
| profit | | |

**(Total for Question 2 = 18 marks)**

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Working and answer

# 3. Question 3 · part a.vi · 2 marks

2021 NOV · Paper 02

Further practice · Attempt part a.vi only.

**3** Dates are often used in programs.

(a) A program accepts the number of a month and outputs the number of days in the month.

Open **Q03a** in the code editor.

Use the code to answer these questions.

(i) Identify the name of a local variable.

(1)

(ii) Identify the name of a data structure.

(1)

(iii) Identify the name of a built-in subprogram.

(1)

(iv) Identify the name of a user-written function.

(1)

(v) Identify the name of a parameter.

(1)

(vi) The program includes validation to ensure the month number is between 1 and 12 inclusive.

Complete the table by adding the type of test data that is being used to test the validation.

The first type of test has been completed for you.

(2)

| Type of test data | Test data |
|---|---|
| Normal | 3 |
| | 1 |
| | 15 |

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# 3. Question 3 · part a.vi · 2 marks

2021 NOV · Paper 02

Further practice · Attempt part a.vi only.

(b) Explain **one** reason why parameters are used to pass data into subprograms.

(2)

(c) A leap year is a year that has 366 days rather than 365 days. A leap year usually occurs every four years.

A year that is a leap year:

- can be exactly divided by 400

Or

- can be exactly divided by 4, but **not** exactly divided by 100.

These years are leap years: 2000, 2012 and 2020.

These years are not leap years: 1700, 1900 and 2021.

A program is needed that will accept the input of a year and output whether it is a leap year or not.

Open **Q03c** in the code editor.

Write the program.

You **must** use the structure given in **Q03c**.

Do not add any further functionality.

Save your code as **Q03cFINISHED** with the correct file extension for the programming language.

(6)

**(Total for Question 3 = 15 marks)**

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## 3. Question 3 · part a.vi · 2 marks · Working and answer

Working and answer

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# 4. Question 1 · part d · 2 marks

2022 JUN · Paper 02

Further practice · Attempt part d only.

**1** Programmers write code to solve problems.

(a) Identify the description of a variable in a computer program.

(1)

- ☒ **A** A value that cannot be used more than once
- ☒ **B** A value that must be input
- ☒ **C** A value that is always used in a calculation
- ☒ **D** A value that can change

(b) Identify the technique that improves the readability of code.

(1)

- ☒ **A** Using indents on every line
- ☒ **B** Using descriptive names for variables
- ☒ **C** Using the correct operators
- ☒ **D** Using suitable data structures

(c) Complete the table by giving an example value for **each** data type.

The first row has been completed for you.

(2)

| Data type | Example value |
|---|---|
| integer | 12 |
| char | |
| real | |

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## 4. Question 1 · part d · 2 marks

2022 JUN · Paper 02

Further practice · Attempt part d only.

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(d) Describe **one** difference between the data used in boundary testing and the data used in erroneous testing.

**(2)**

(e) A program should output the value 2

However, there is an error in the code and the actual output is 4

Name this type of error.

**(1)**

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(f) Open **Q01f** in the code editor.

The program should allow the input of the length of the side of a square and output the area of the square.

There are **three** errors in the code.

Amend the code to correct the errors.

Save your amended code as **Q01fFINISHED** with the correct file extension for the programming language.

**(3)**

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(g) A program is needed that will accept the input of a letter and compare it with a stored letter.

It will check whether the letter comes earlier in the alphabet, later in the alphabet or is the same letter as the stored letter.

It will output the letter and the stored letter with an appropriate message.

Open **Q01g** in the code editor.

Amend the code to complete the if statement used to produce the output.

You must use the structure given in **Q01g** to complete the program.

Do not add any further functionality.

Save your code as **Q01gFINISHED** with the correct file extension for the programming language.

**(4)**

**(Total for Question 1 = 14 marks)**

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## 4. Question 1 · part d · 2 marks · Working and answer

Working and answer

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# 5. Question 2 · part b.i · 3 marks

2023 JUN · Paper 02

Further practice · Attempt part b.i only.

**2** Solutions to problems are made up of many different components.

(a) Programmers use subprograms when developing code.

(i) Give **two** reasons to use **subprogram libraries** when developing code.

**(2)**

1 .....

.....

2 .....

.....

(ii) Identify which **one** of these must return a result.

**(1)**

- ☒ **A** Function
- ☒ **B** Iteration
- ☒ **C** Procedure
- ☒ **D** Selection

(iii) State what is meant by the term **local variable**.

**(1)**

.....

.....

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## 5. Question 2 · part b.i · 3 marks

2023 JUN · Paper 02

Further practice · Attempt part b.i only.

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(b) A school is planning a fish and chip dinner for the students and their families.

(i) Tickets are only sold to families.

A family is one to four adults and one to six children.

A program is being written to help manage ticket sales.

Complete the table to show examples of normal, boundary and erroneous numeric test data for the program.

(3)

| | Adult | Children |
|---|---|---|
| Normal | | |
| Boundary | | |
| Erroneous | | |

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(ii) The kitchen staff use a program to determine whether they have enough chips or how many more they need to order.

Open **Q02bii** in the code editor.

There are **four** errors in the code.

Amend the code to correct the errors.

Use this test data to help you find the errors.

| Chips in stock (kilograms) | Number of adults | Number of children | Expected output |
|---|---|---|---|
| 19 | 100 | 150 | Stocks available |
| 15 | 100 | 150 | Order 4.0 kilograms |

Save your amended code as **Q02biiFINISHED** with the correct file extension for the programming language.

(4)

**(Total for Question 2 = 11 marks)**

## 5. Question 2 · part b.i · 3 marks · Working and answer

Working and answer

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# 6. Question 2 · part d · 4 marks

2024 JUN · Paper 02

Further practice · Attempt part d only.

**2** Programmers design, correct and test programs.

(a) Programmers design programs.

(i) Identify the term for a step-by-step description to complete a task.

(1)

- ☒ **A** Algorithm
- ☒ **B** Computational thinking
- ☒ **C** Decomposition
- ☒ **D** Pseudocode

(ii) A constant is a memory location whose value does not change during program execution.

Give the name for a memory location whose value can change during program execution.

(1)

(b) Programs can have runtime errors.

(i) State what is meant by the term **runtime error**.

(1)

(ii) Give **one** example of a runtime error.

(1)

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# 6. Question 2 · part d · 4 marks

2024 JUN · Paper 02

Further practice · Attempt part d only.

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(c) Open **Q02c** in the code editor.

The program generates a random number for the width of a rectangle.

The length of the rectangle is always 4

The program should calculate the perimeter of the rectangle.

The program should display the width, the length and the perimeter of the rectangle.

There are **three** errors in the code.

Amend the code to correct the errors.

Save your amended code as **Q02cFINISHED** with the correct file extension for the programming language.

**(3)**

(d) **Figure 1** shows part of an online form displayed in a browser.

| | | |
|---|---|---|
| Memorable word | | (2-8 letters) |
| Date | 31-01-2019 | |

**Figure 1**

Complete the table to show **two** examples of erroneous, **one** example of normal and **one** example of boundary test data for the **memorable word** field.

Each example must be different.

**(4)**

| | |
|---|---|
| **Erroneous** | |
| **Erroneous** | |
| **Normal** | |
| **Boundary** | |

**(Total for Question 2 = 11 marks)**

## 6. Question 2 · part d · 4 marks · Working and answer

Working and answer

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